

SOOM — API Client Walkthrough

Step-by-step visual reference of the use case, the end results, and the complete functionality of the API client using YouTube API Services.

API Client	SOOM (숨) — filming-location service for video creators
Live URL	https://gwanjetap.netlify.app
GCP project number	507307814114 (project travel-505308)
Requested quota	300,000 units/day (current default: 10,000)
Document date	13 August 2026
Interface language	Korean. Every screen below is translated in this document.

1. What the client does, in one paragraph

SOOM helps video creators in Korea find places to film in depopulating rural regions. A creator enters their own YouTube channel. We look up that channel's public statistics, work out which *subjects* the channel actually performs well on (traditional five-day markets, closed schools, islands, valleys...), and then list public-data locations across Korea where that same subject can be filmed — ranked so that places with little existing video coverage surface first.

YouTube API Services supply exactly two things: **public channel statistics** for the creator who is asking, and **public video statistics** used to compute an aggregate performance score per subject. Everything else on screen comes from Korean public open data (the national tourism API, traditional-market registry, closed-school registry, sunrise/sunset tables).

No end-user ever triggers a YouTube API call. All collection runs server-side in scheduled batch scripts and lands in our own database. The web front end reads a JSON snapshot baked at build time, so visitor traffic consumes zero quota. This is verified in step 9.

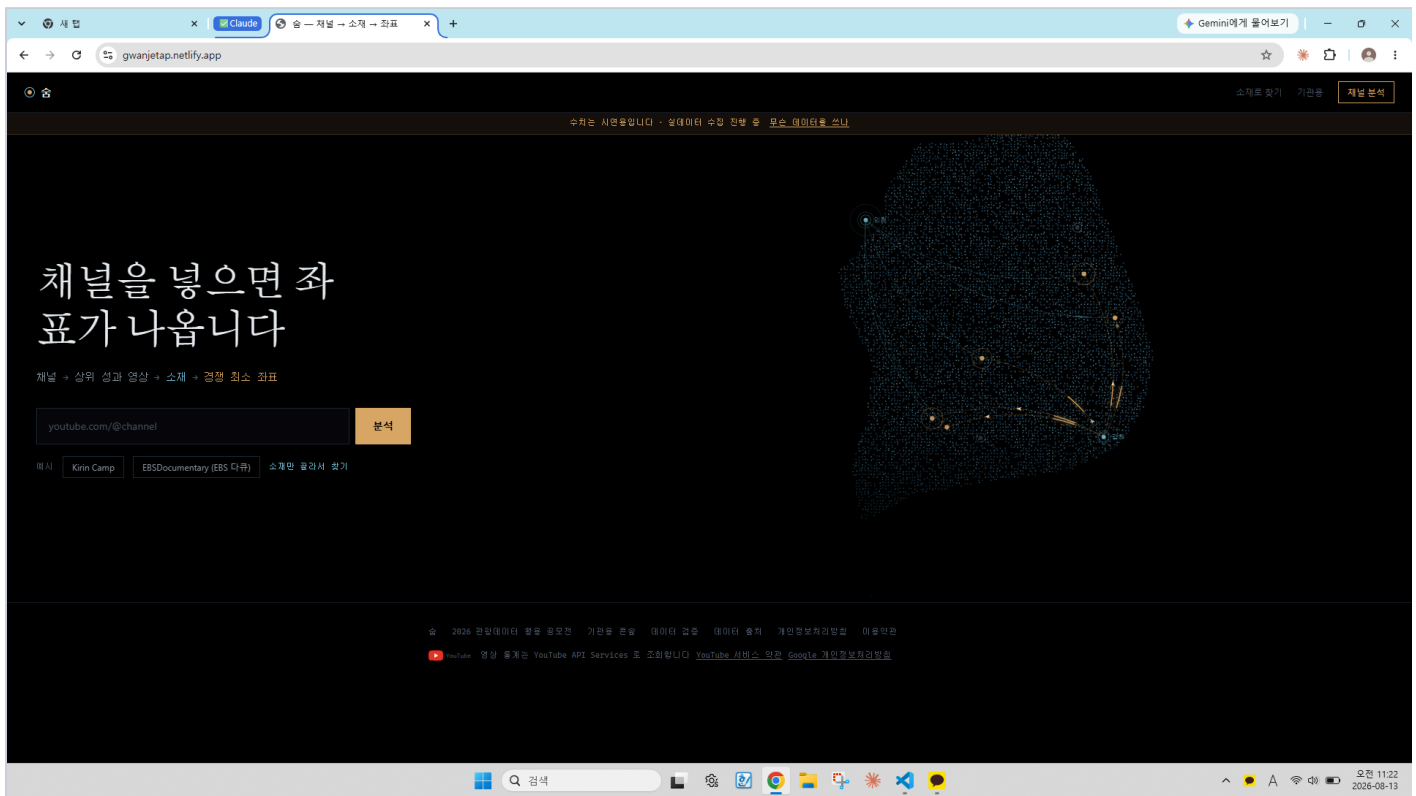
2. Where each YouTube API call is used

API call	Purpose in the product	Seen in step
channels.list	Subscriber count and uploads playlist id for the channel the creator entered.	Step 2
playlistItems.list	Recent uploads of that channel (used instead of search.list — 1 unit rather than 100).	Step 2
videos.list	View count, duration, description (for chapter timestamps), 50 ids per call.	Steps 2, 3, 5
search.list	Discover candidate videos per subject and per region name. This is the call the quota increase is for.	Steps 3, 5
commentThreads.list	Identify the filming location named by viewers when the title does not name it.	Step 5 (attribution)

3. Step-by-step walkthrough

STEP 1 Entry point — the creator enters a YouTube channel

`no API call yet` The headline reads "Enter a channel and you get coordinates". The input accepts a channel URL, handle, or name. The two buttons below are example channels. The footer carries YouTube branding and links to the YouTube Terms of Service and the Google Privacy Policy.



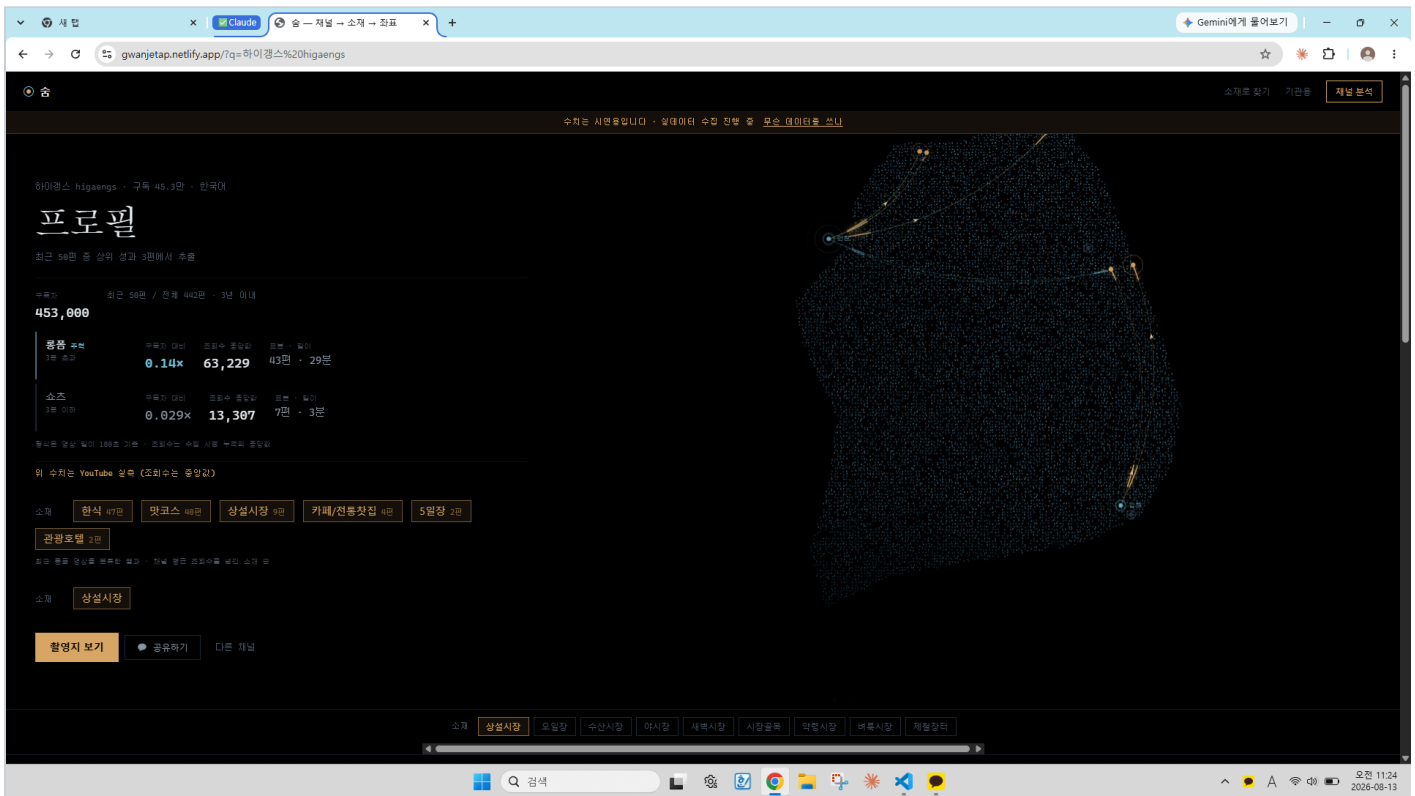
<https://gwanjetap.netlify.app/>

STEP 2 Channel analysis — the end result of `channels.list` + `playlistItems.list` + `videos.list`

`channels.list` `playlistItems.list` `videos.list` The creator's channel is resolved and its public statistics are shown. Reading the screen top to bottom:

- **453,000** — subscriber count (`channels.list`).
- **Long-form 0.14× / 63,229** — median views ÷ subscribers, and median view count, across the 43 long-form uploads in the sample of 50 (`videos.list`).
- **Shorts 0.029× / 13,307** — the same split for the 7 uploads under 180 seconds.
- The line under the numbers states plainly: *"the figures above are measured from YouTube (view counts are medians)."*
- The chips are the subjects this channel actually films, derived by classifying its own video titles and descriptions: Korean food 47, food routes 40, permanent markets 9, cafés 4, five-day markets 2, hotels 2.

We normalise by subscriber count rather than raw views. A 6.2M-subscriber channel in our sample scores 0.016× while a 2,080-subscriber channel scores 1.539× — the ranking inverts once channel size is removed. This is the core reason we need channel statistics at all.



<https://gwanjetap.netlify.app/?q=하이갱스 higaengs>

STEP 3 Subject evidence — aggregate score and the actual videos behind it

`search.list` `videos.list` The creator picked the subject "five-day market". This block is the end result of the collection the quota increase pays for.

- **122,763 ~ 847,563** — projected reach for this channel on this subject, always shown as a *range* with the caption "estimate based on past data", never as a single number.
- The three cards are **real YouTube videos** embedded through the official youtube-nocookie.com iframe player, each with a "View on YouTube ↗" link back to the original watch page. The figures under each card are the measured ones: 16.3× (1,800 subscribers → 30,000 views), 6.6× (6,200 → 41,000), 6.1× (6,500 → 40,000).
- The bar underneath reads "already filmed" and names the places that do have coverage — Namhae 3, Goryeong 3, Mungyeong 3 — so that "0 competing videos" on the cards below has a reference point.
- The results header states the provenance: "from 458 public-data locations · real data".

The leftmost card shows YouTube's own "Video unavailable — the owner has disabled playback on other websites" message. That is the official player enforcing the uploader's embedding setting, and we deliberately do not work around it: the card still credits the video and links to its watch page.

The screenshot shows a web application interface with a search bar at the top. Below the search bar, there are three video cards. The first card on the left shows a YouTube video player with a message: "동영상 재생할 수 없음 동영상 소유자가 다른 웹사이트에서 재생할 수 있도록 설정했습니다. YouTube에서 보기". The second card in the middle shows a video titled "진안 오일장(전통시장)" with a view count of 16.3x. The third card on the right shows a video titled "안가보면 후회하는" with a view count of 6.1x. Below the video cards, there is a section titled "결과" (Results) which lists three locations: 대외시장 (4.9일), 동충전통시장, and 봉평재래시장 (2.7일). Each location has a small image and some statistics. At the bottom of the page, there is a search bar and some navigation links.

https://gwanjetap.netlify.app/?q=하이갠스 higaengs&tag=t_oil_market#result

STEP 4 A recommended location, expanded

public open data only Expanding a card shows what a creator needs in order to actually go: the next market days with civil-twilight, sunrise and sunset times (from the Korean tourism and astronomy public datasets), other subjects within driving distance, and the number of competing videos for each. No YouTube data on this panel except the competing-video counts.

The screenshot shows a web application interface for market information. The browser address bar displays the URL: `https://gwanjetap.netlify.app/?q=하이겐스%20higaengs&tag=t_oil_market#result`. The application has a dark theme and a navigation bar at the top with tabs: "소계", "요즘장", "수산물시장", "상물시장", "야시장", "새벽시장", "시장골목", "작당시장", "버룩시장", and "계동장터".

The main content area features a table of market days:

다음 장날	4월+9일	장료	49
8월 14일 (금)	비명 05:11 일출 05:39 일몰 19:19		
8월 19일 (수)	비명 05:15 일출 05:43 일몰 19:12		
8월 24일 (월)	비명 05:20 일출 05:47 일몰 19:06		

Below the table, there is a section titled "근처 6인 근처" (Nearby 6 people nearby) with a list of nearby markets:

- 금당계곡 5.7km · 차 9분 · 계곡 · 경정 1면
- 술삼계곡 5.7km · 차 18분 · 야당계곡 · 경정 0면
- 장전계곡 8.3km · 차 12분 · 계곡 · 경정 1면
- 관전대 10.6km · 차 16분 · 유목지 · 경정 0면

At the bottom, there are two market cards with photos and details:

- 동송전동시장** 강원특별자치도 철원군 [장무함소]
오일장 영상 0.586x · 경정 0면
장구 84면 · 기온 95% 0.491~0.847 · 구목지 구안 합산
구정 보기
- 봉평재래시장 (2, 7일)** 강원특별자치도 평원군 [장무함소]
오일장 영상 0.586x · 경정 0면
장구 84면 · 기온 95% 0.491~0.847 · 구목지 구안 합산
구정 보기

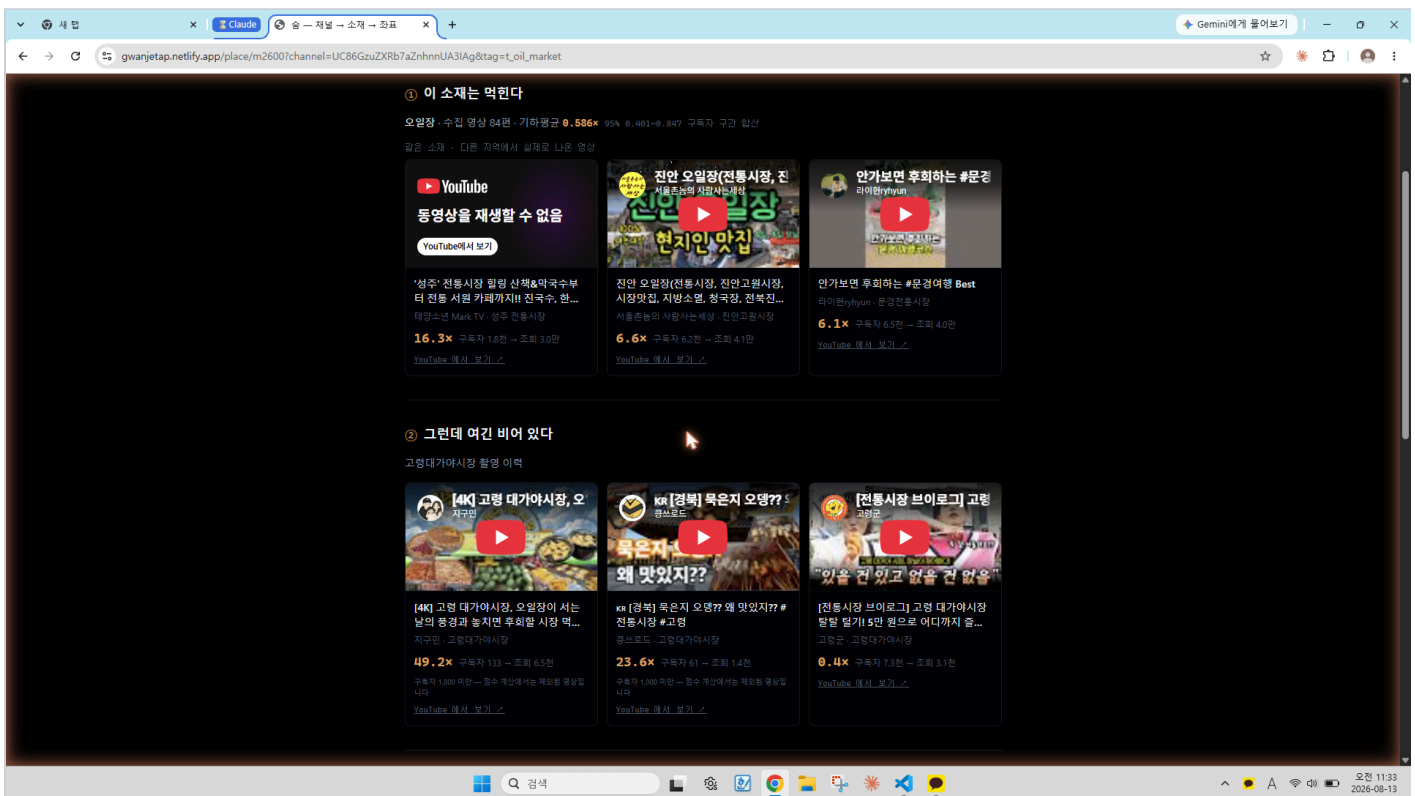
https://gwanjetap.netlify.app/?q=하이겐스 higaengs&tag=t_oil_market#result

STEP 5 Location detail, evidence ① and ② — six real embedded players

`search.list` `videos.list` `commentThreads.list` This is the deepest screen and the one where YouTube data is most visible.

- ① **"This subject works"** — the aggregate: *five-day markets* · 84 collected videos · geometric mean $0.586 \times$ · 95% CI 0.401–0.847. Below it, three real videos of the same subject *filmed elsewhere*, each embedded and linked back.
- ② **"But this place is empty"** — every video we could attribute to *this specific location*. Here there are three: $49.2 \times$ (133 subscribers → 6,500 views), $23.6 \times$ (61 → 1,400), $0.4 \times$ (7,300 → 3,100). The two small-channel cards carry the note "under 1,000 subscribers — excluded from score calculation", because a tiny denominator makes the ratio explode.

Attributing a video to a location is what `commentThreads.list` is for. Korean travel videos frequently name the place only in the comments, and English-language ones usually do not name it in the title at all.



https://gwanjetap.netlify.app/place/m2600?channel=UC86GzuZXRb7aZnhnnUA3IAg&tag=t_oil_market

STEP 6 Evidence ③ and ④ — why the place is empty, and projected reach

videos.list ③ answers "is it empty because it is bad?" with public-data facts: it is registered with the Korea Tourism Organization; there is no public-data photo; the 13 locations of this subject that do have coverage average 1.5 videos while this one has 3; nationally 244 of 458 five-day markets are in officially designated depopulating districts.

④ is the projected reach for this specific channel, again as a range (**122,763 ~ 847,563**) with the disclaimer "estimate based on past data". We never publish a single predicted view count.

The screenshot shows a web browser window with the address bar displaying `gwanjetap.netlify.app/place/m2600?channel=UC86GzuZXRb7aZnhnnUA3IAg&tag=t_oil_market`. The page content is in Korean and includes the following sections:

- ③ 별로라서가 아니다**
 - 한국관광공사 등록 정보 있음
 - 공공데이터 사진 없음
 - 같은 소재에서 영상이 잡힌 곳 13곳의 평균 1.5편, 여긴 3편
 - 전국 오일장 458곳 중 인구감소지역 244곳
- ④ 당신이면 이 정도**
 - 구독 45.3만 기준
 - 122,763 ~ 847,563**
 - 과거 데이터 기반 추정
- ⑤ 이렇게 찍으면 된다**

다음 장날	요일	시간	종료
8월 14일 (금)	박영 05:16	김종 05:43	김영 19:17
8월 19일 (수)	박영 05:20	김종 05:47	김영 19:11
8월 24일 (월)	박영 05:24	김종 05:51	김영 19:05

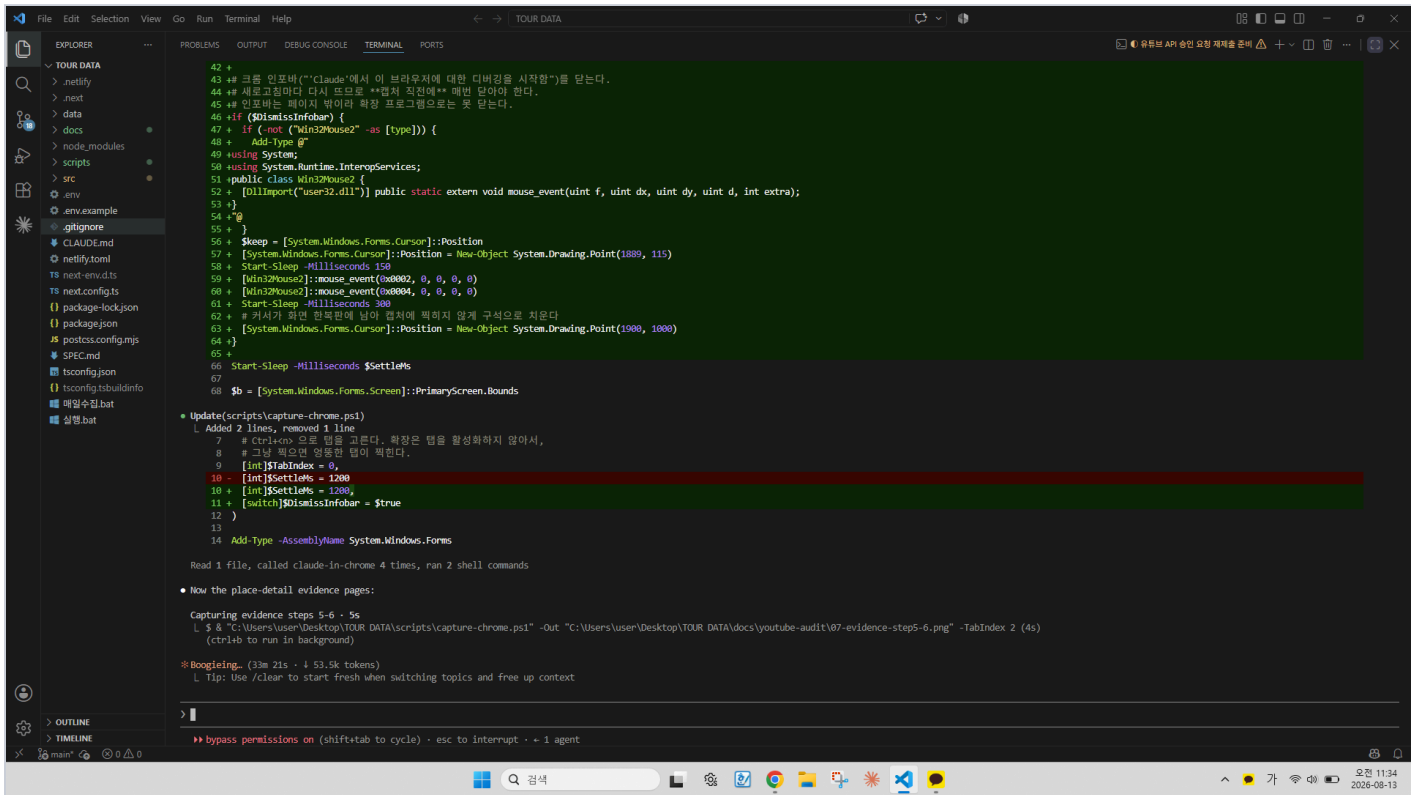
장날 : 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장, 관동시장
- ⑥ 이렇게 머물면 된다**
 - 근처에 같이 찍을 소재
 - 고령 자산리 당간지주, 사당
 - 거리 5분, 경정 0편

https://gwanjetap.netlify.app/place/m2600?channel=UC86GzuZXRb7aZnhnnUA3IAg&tag=t_oil_market

STEP 7 Evidence ⑤ and ⑥ — when to shoot, what else is nearby

public open data only

The market opens on days ending in 4 and 9; the next three dates are listed with civil twilight, sunrise and sunset for each, sourced from the Korea Astronomy and Space Science Institute. Below, six other filmable subjects within a 40 km radius, each with its competing-video count. The provenance line is printed under each block.



```
42 +
43 + # 코롬 인포바 ("Claude"에서 이 브라우저에 대한 디버깅을 시작함)를 받는다.
44 + # 새로 고침마다 다시 프롬프트 **캡처 직전에** 매번 닫아야 한다.
45 + # 인포바는 페이지 위아래 확장 프로그램으로는 못 받는다.
46 + if ($?IsMissInfoBar) {
47 + if (-not ("Win32Mouse2" -as [Type])) {
48 + Add-Type @"
49 + using System;
50 + using System.Runtime.InteropServices;
51 + public class Win32Mouse2 {
52 + [DllImport("user32.dll")] public static extern void mouse_event(uint f, uint dx, uint dy, uint d, int extra);
53 + }
54 + #
55 + }
56 + $keep = [System.Windows.Forms.Cursor]::Position
57 + [System.Windows.Forms.Cursor]::Position = New-Object System.Drawing.Point(1889, 115)
58 + Start-Sleep -Milliseconds 150
59 + [Win32Mouse2]::mouse_event(0x0002, 0, 0, 0, 0)
60 + [Win32Mouse2]::mouse_event(0x0004, 0, 0, 0, 0)
61 + Start-Sleep -Milliseconds 300
62 + # 커서가 화면 왼쪽판에 남아 캡처에 찍히지 않게 구석으로 치운다
63 + [System.Windows.Forms.Cursor]::Position = New-Object System.Drawing.Point(1980, 1880)
64 + }
65 + }
66 Start-Sleep -Milliseconds $SettleMs
67
68 $b = [System.Windows.Forms.Screen]::PrimaryScreen.Bounds
69
70 Update-Scripts\capture-chrome.ps1
71 L Added 2 lines, removed 1 line
72 # Curl-Ext으로 캡처 고른다. 확장은 텀을 활성화하지 않아서,
73 # 그냥 찍으면 엉뚱한 텀이 찍힌다.
74 [int]$TabIndex = 0,
75 [int]$SettleMs = 1200
76 [int]$SettleMs = 1200,
77 [switch]$IsMissInfoBar = $true
78
79 Add-Type -AssemblyName System.Windows.Forms
80
81 Read 1 file, called claude-in-chrome 4 times, ran 2 shell commands
82
83 New the place-detail evidence pages:
84 Capturing evidence steps 5-6 - 5s
85 L $ & "C:\Users\user\Desktop\TOUR DATA\scripts\capture-chrome.ps1" -Out "C:\Users\user\Desktop\TOUR DATA\docs\youtube-audit\07-evidence-steps5-6.png" -TabIndex 2 (4s)
86 (ctrl+b to run in background)
87
88 Boogieing... (33s 21s - 4 53.5k tokens)
89 L Tip: Use /clear to start fresh when switching topics and free up context
90
91 >
92
93 >> bypass permissions on (shift+tab to cycle) - esc to interrupt - + 1 agent
```

https://gwanjetap.netlify.app/place/m2600?channel=UC86GzuZXRb7aZnhnnUA3IAg&tag=t_oil_market

STEP 8 Attribution page — /data-sources

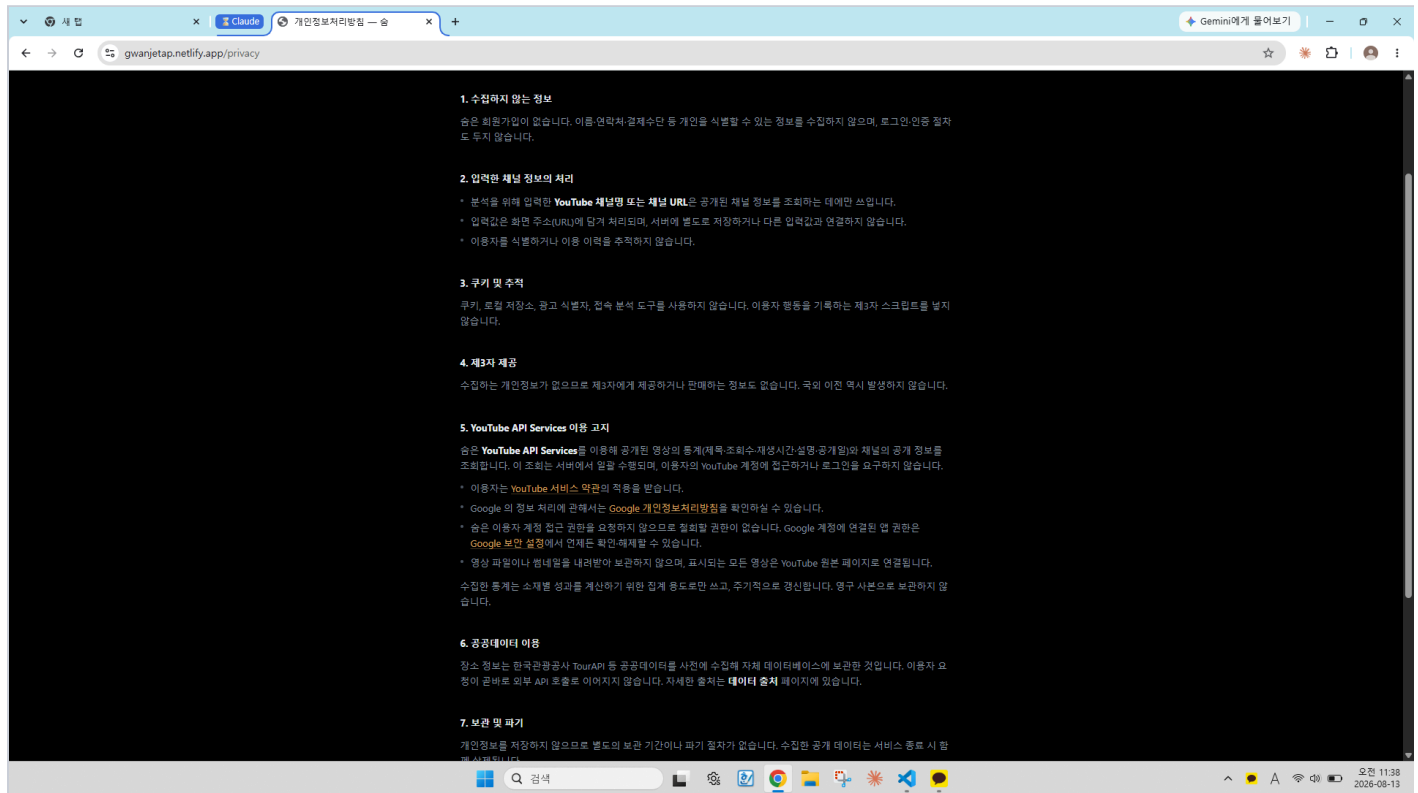
Linked from the footer of every page. It lists every data source including "YouTube Data API v3 — Google", explains the scoring method, and demonstrates the official embedded player with a link to the watch page. The section headed "What we do not do" states: we do not download or store video files or thumbnails; the front end makes no direct external API calls; projected reach is never a single number; there is no sign-up, login, advertising or behavioural tracking. The page links to the YouTube Terms of Service and the Google Privacy Policy.



<https://gwanjetap.netlify.app/data-sources>

STEP 9 Privacy policy — YouTube API Services disclosure

Section 5 is the required disclosure. It states that SOOM uses YouTube API Services to retrieve public video statistics (title, view count, duration, description, publish date) and public channel information; that retrieval is performed server-side in batches and never asks the user to log in or grants us access to their YouTube account; that users are bound by the YouTube Terms of Service; that Google's handling of information is covered by the Google Privacy Policy; that because we request no user account access there is nothing to revoke, with a link to Google security settings; and that we retain no video files or thumbnails, refreshing collected statistics periodically rather than keeping a permanent copy. Sections 1–4 confirm there is no sign-up, no cookies, no tracking and no third-party sharing.



<https://gwanjetap.netlify.app/privacy>

4. How collection actually runs (server-side only)

Collection is a set of Node.js command-line scripts run on a schedule, never from the browser. Each run writes its unit consumption to a `yt_quota` table keyed by date, because a process that forgets consumption restarts the count from zero on every re-run and overshoots the daily cap.

Command	Calls used	Cost
<code>npm run yt:channel -- @handle</code>	<code>channels.list</code> → <code>playlistItems.list</code> → <code>videos.list</code>	3 units per 50 videos
<code>npm run yt:search</code>	<code>search.list</code> → <code>videos.list</code>	100 units per page
<code>npm run yt:comments</code>	<code>commentThreads.list</code>	1 unit per video

Current contents of our database, collected entirely within the default 10,000 units/day:

Table	Rows
Videos (public metadata only)	42,018
Channels (public statistics only)	2,542
Comments (used for location attribution)	21,867
Video → location attributions	2,978
Filming locations from Korean public open data	50,033

We deliberately avoid `search.list` wherever a cheaper call exists. Fetching a channel's recent uploads via `search.list` costs 100 units; going through `channels.list` → `playlistItems.list` costs 3. Measured on our own data, that is 50 videos for 3 units. The quota increase is needed only for the one job that genuinely requires search: discovering candidate videos for subjects and regions we have never seen.

5. Why 300,000 units/day

`search.list` costs 100 units per page. Under the default 10,000/day that is 100 searches per day. The initial corpus needs roughly 700 distinct queries (229 Korean municipalities × 3 phrasings, 66 subjects, ~30 English queries) at 3 pages each — about 210,000 units — which would take over a month of uninterrupted collection with no margin for retries.

Scores must be computed separately per language. In our sample, traditional markets score 4.1× for English-language videos and 0.6× for Korean-language ones; merging the two collapses every subject toward the mean and makes the recommendation meaningless. Both language corpora therefore have to be collected rather than sampled. Our English-language corpus is currently the gap: enter an English-language channel today and the screen honestly reports "rank only — insufficient sample".

At 300,000 units/day the initial collection completes in about two days, after which ongoing use falls back to roughly 10,000 units/day for periodic re-scoring.

6. Compliance summary

- No YouTube content is downloaded, stored as media, or re-hosted. Only public metadata.

- Every video is played through the official [youtube-nocookie.com](https://www.youtube-nocookie.com) iframe player, and every card carries a link back to the watch page. Uploader embedding settings are respected, not circumvented.
- Stored statistics are refreshed periodically rather than retained indefinitely as a static copy.
- Aggregate figures are shown with sample size and a 95% confidence interval; multipliers are printed only when the sample reaches 100 videos, otherwise the screen says "rank only".
- Projected reach is always a range with an explicit estimate disclaimer, never a guaranteed outcome.
- No user authentication, no OAuth, no access to private user data, no advertising, no behavioural tracking, no cookies.
- The API key is server-side only and is never exposed to the browser; the front end makes no direct calls to the YouTube API.
- Screens that still contain illustrative rather than collected data are labelled as such on the page itself, in the interface and on [/data-sources](#).